INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

SOFTWARE ENGINEERING ASSIGNMENT REPORT

Nepal Disaster Management Case Study

Figure 1. Nepal flood disaster impact — photograph supplied for the assignment.

Submitted by

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Course: Software Engineering
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TEAM CONTRIBUTION

The following table presents the team contribution for the Intelligent Disaster Early Warning and Evacuation Management System project. The responsibilities are organized around the major activities covered in the report, including requirements, design, development, testing, documentation and validation.

Team Member	Register No.	Contribution / Responsibility	Approx. Contribution
R KALAI AMUDHAN	192511292	Problem formulation, requirements engineering, stakeholder needs and UML/use-case analysis.	12%
SHAKTHIVEL	192524325	Backend structure, authentication, database design and application-service development.	13%
MURALI KRISHNAN N	192524090	Monitoring inputs, data validation, risk assessment logic and algorithm development.	13%
RR THIYAGESH	192572007	Alert generation, communication workflow, authority dashboard and integration support.	13%
MUKESH RAJ L	192571036	GIS evacuation workflow, shelter management, route analysis, testing and validation support.	14%
Jerold Samuel J	192571052	Citizen reporting workflow, interface integration and user-oriented functionality.	12%
S RUPESH	192571055	System testing, security considerations, test-case analysis and result verification.	11%
Divya Prashanth S	192521361	Technical documentation, SDG/ethical considerations, references and final report coordination.	12%

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1. Problem Statement and Problem Formulation

Nepal faces hazards including floods, landslides, earthquakes and severe weather. Disaster impacts can be worsened by delayed warnings, fragmented information, blocked transport routes and limited visibility of shelter capacity and emergency resources.

Figure 2. Emergency response activity during flooding — photograph supplied for the assignment.

The software problem is to provide a centralized platform that combines monitoring data, risk assessment, location-specific warning, evacuation assistance, shelter management, citizen reporting and authority coordination.

Problem formulation: design a secure, reliable, maintainable and scalable software system that receives disaster observations, evaluates risk, generates warnings and supports evacuation and emergency decision-making.

2. Objective and Expected Outcomes

- Develop a centralized disaster monitoring and early-warning platform.
- Collect and validate IoT, weather and environmental observations.
- Assess risk using explainable rules with an architecture ready for ML integration.
- Generate location-specific multi-channel alerts.
- Provide GIS-based evacuation routes and nearby shelter information.
- Monitor shelter capacity, facilities and emergency resources.
- Enable citizen incident reporting.

Expected Outcomes

- Faster detection and communication of hazards.

- Improved citizen–authority communication.
- Better-organized evacuation and shelter allocation.
- Improved emergency resource visibility.
- A modular prototype suitable for future live-data and ML integration.

3. Requirements, Constraints and Assumptions

3.1 Functional Requirements

ID	Requirement	Priority
FR01	Authentication and role-based access	Must Have
FR02	Receive and validate monitoring data	Must Have
FR03	Calculate and classify risk	Must Have
FR04	Create and distribute alerts	Must Have
FR05	Display affected areas and evacuation routes	Must Have
FR06	Locate shelters and availability	Must Have
FR07	Citizen incident reporting	Should Have
FR08	Authority shelter/resource updates	Should Have
FR09	Historical reports and analytics	Could Have

3.2 Non-Functional Requirements

Attribute	Requirement
Security	Authentication, authorization, validation, password hashing and audit logging.
Reliability	Error handling, input validation and fallback mechanisms.
Usability	Clear warnings and simple evacuation instructions.
Performance	Routine dashboard/API operations should respond within a few seconds.
Scalability	Support growth in users, locations and data sources.
Maintainability	Modular code, documentation and reusable services.

3.3 Constraints and Assumptions

- Prototype sensor feeds may be simulated.
- Live deployment requires authorized data sources.
- Network connectivity may fail during emergencies.
- GIS route quality depends on current road information.
- ML quality depends on appropriate historical and labelled data.
- Emergency authorities remain responsible for operational decisions.

4. Application of Relevant Software Engineering Concepts

Concept	Application
Requirements Engineering	Stakeholders, use cases, functional/non-functional requirements and prioritization.
Agile / Scrum	Product backlog, sprint planning, reviews and iterative development.
UML	Use-case, class, activity, sequence and component models.
Modular Architecture	Monitoring, risk, alert, GIS, shelter and citizen modules.
MVC	Separates presentation, business logic and data access.
Observer Pattern	Risk changes notify registered alert channels.
Strategy Pattern	Evacuation route strategy can vary by disaster context.
Testing	Unit, integration, system, black-box and acceptance testing.
Risk Management	Technical, operational, security and data risks with mitigations.
Configuration Management	Git branches, commits, versioning and controlled releases.

5. Design / Proposed Solution / Methodology

5.1 System Workflow

1. Collect observations from sensors or simulated sources.
2. Validate incoming readings.
3. Calculate a risk score.
4. Classify risk as Low, Moderate, High or Critical.
5. Generate an alert when the configured threshold is exceeded.
6. Identify affected geographic areas.
7. Find feasible evacuation routes and available shelters.
8. Notify relevant citizens and authorities.
9. Store events, alerts, actions and reports for analysis.

5.2 Four Main Modules

Module	Description
1. Monitoring & Risk	Collects observations, validates data and determines risk.
2. Alert & Communication	Creates targeted warnings and distributes them through available channels.
3. GIS Evacuation & Shelter	Finds feasible routes and shelters using current conditions.
4. Dashboard & Citizen Reporting	Provides authority monitoring and two-way incident reporting.

5.3 Agile Development Plan

Sprint	Deliverables
1	Requirements, user stories, database and authentication.
2	Monitoring interface and risk engine.
3	Alert service and authority dashboard.
4	GIS evacuation and shelter management.
5	Citizen reporting, integration and testing.

6. System Architecture and Design

A three-tier MVC-oriented architecture is proposed. The presentation layer communicates with application services, while the data layer stores operational and historical information. External monitoring, GIS and communication services are connected through APIs.

Layer	Components	Responsibility
Presentation	Citizen UI, Authority Dashboard, GIS Map	Displays alerts, maps, reports and status.
Application	Auth, Risk, Alert, Evacuation, Shelter Services	Implements business rules and workflows.
Data	MySQL entities for users, sensors, events, alerts, shelters and reports	Stores current and historical data.
External	IoT, weather, GIS and communication APIs	Provides monitoring and communication services.

Security Design

- Role-Based Access Control for Citizen, Authority, Emergency Team and Administrator.
- Password hashing and secure session/token handling.
- Input validation for sensor readings and reports.
- Audit logs for sensitive authority operations.

7. UML and Software Design Models

7.1 Use-Case Model

Actors: Citizen, Authority, Emergency Team and Administrator. Major use cases: Login, View Alert, View Map, Find Shelter, Follow Route, Report Incident, Monitor Risk, Manage Alerts, Manage Shelters, Update Resources and Manage Users.

7.2 Class Model

Class	Attributes	Methods
User	userId, name, role, email	login(), logout()
SensorData	sensorId, location, value, timestamp	validate()
DisasterEvent	eventId, type, severity, location	updateStatus()
Alert	alertId, level, message, area	publish()
Shelter	shelterId, location, capacity, occupancy	updateCapacity()
IncidentReport	reportId, citizenId, location, severity	submit()
EvacuationRoute	routeId, source, destination, distance	calculate()

7.3 Sequence / Activity / Component Models

Sequence: Monitoring Source → API → Validation → Risk Engine → Alert Service → Database → Dashboard / Citizen Notification.

Activity: Critical Alert → Identify Area → Locate Shelters → Check Road Status → Calculate Route → Display Route → Evacuate → Monitor.

Components: Sensor Gateway, Risk Service, Alert Service, GIS Service, Shelter Service, Citizen Service, Dashboard Service and Database.

8. Algorithm / Pseudocode / Flowchart

8.1 Risk Assessment

```
START
Receive rainfall, water-level, wind/seismic/environment readings.
Validate inputs and normalize selected indicators.
Calculate risk score.
IF score >= 81: CRITICAL
ELSE IF score >= 61: HIGH
ELSE IF score >= 31: MODERATE
ELSE: LOW
Store result.
IF HIGH or CRITICAL: generate alert.
END
```

8.2 Evacuation Route

```
START
Receive current location.
Find nearby shelters.
Remove blocked/unsafe roads from route graph.
Calculate candidate routes.
Compare distance, safety and shelter availability.
Select best feasible route.
Display route on GIS map.
END
```

8.3 Example Function

```
def classify_risk(score):
    if score >= 81: return "CRITICAL"
    elif score >= 61: return "HIGH"
    elif score >= 31: return "MODERATE"
    return "LOW"
```

9. Implementation / Source Code / Environment and Tools

Area	Technology / Tool
Frontend	HTML5, CSS3, JavaScript
Backend	Python Flask
Database	MySQL
Data Processing	Python, Pandas, NumPy
ML Extension	Scikit-learn
GIS	Leaflet.js + OpenStreetMap
API Testing	Postman
IDE	Visual Studio Code
Version Control	Git / GitHub

Suggested Project Structure

```
intelligent-disaster-system/
├── app.py
├── config.py
├── models/
├── routes/
├── services/
├── templates/
├── static/
├── tests/
├── dataset/
└── requirements.txt
```

10. Test Cases and Expected / Actual Results

ID	Scenario	Expected	Actual	Status
TC01	Valid login	Dashboard opens	Dashboard opens	PASS
TC02	Invalid password	Access rejected	Access rejected	PASS
TC03	High-risk readings	HIGH/CRITICAL shown	Risk shown	PASS
TC04	Critical event	Emergency alert created	Alert created	PASS
TC05	Shelter full	Alternative shelter shown	Alternative shown	PASS
TC06	Blocked road	Unsafe route excluded	Route recalculated	PASS
TC07	Citizen report	Stored and visible	Stored and visible	PASS
TC08	Unauthorized admin page	Access denied	Access denied	PASS

Testing Levels

- Unit testing – individual functions.
- Integration testing – module/database interaction.
- System testing – complete disaster workflow.
- Black-box testing – input/output behaviour.
- User acceptance testing – representative citizen and authority workflows.

11. DISASTER PHOTOGRAPHS

The following Nepal disaster photographs provide the real-world case-study context. For the final software demonstration, insert screenshots of the implemented login page, risk dashboard, alert screen, GIS route, shelter status and citizen-report form.

Figure 3. Flood-affected settlement and river corridor — photograph supplied for the assignment.

12. Results and Validation

Scenario	Input	System Response	Validation
Normal	Low readings	LOW risk	Correct
Heavy Rain	Increasing rainfall	MODERATE/HIGH warning	Correct
Flood	Rapid water-level rise	HIGH/CRITICAL + evacuation support	Correct
Blocked Road	Road marked unsafe	Alternative route	Correct
Shelter Full	Capacity reached	Alternative shelter	Correct
Citizen Incident	Report submitted	Stored + visible to authority	Correct

Performance Metrics

- Average API response time.
- Alert-generation latency.
- Risk-classification accuracy on a validated dataset.
- Percentage of successful test cases.
- Database query response time.
- Route computation time.

13. Analysis, Comparison, Trade-offs and Justification

Approach	Advantage	Trade-off	Decision
Rule-Based Risk	Transparent and explainable	Less adaptive than ML	Prototype baseline
ML Risk Model	Can learn complex patterns	Needs quality data	Future extension
Central Dashboard	Single operational view	Central dependency	Selected
Cloud Deployment	Scalable	Cost/security requirements	Recommended
Dijkstra Routing	Simple and explainable	Needs accurate road graph	Prototype
Multi-Channel Alerts	Improves reach	External dependencies	Selected

The selected architecture balances transparency, maintainability, safety and extensibility. Rule-based classification is appropriate for a prototype because its decisions can be explained and tested, while the architecture allows future ML and live-data integration.

14. Broader Considerations / SDG Relevance

SDG	Relevance
SDG 11 – Sustainable Cities and Communities	Supports resilient communities, preparedness and evacuation planning.
SDG 13 – Climate Action	Supports disaster-risk preparedness and climate adaptation.
SDG 9 – Industry, Innovation and Infrastructure	Uses digital infrastructure and software innovation for public safety.
SDG 3 – Good Health and Well-being	Supports emergency access and protection of life.

Ethical and Privacy Considerations

- Collect personal/location data only when necessary.
- Protect citizen reports and sensitive emergency information.
- Treat AI predictions as decision support, not absolute truth.
- Use verified warnings to reduce misinformation and panic.
- Design for vulnerable groups and local-language communication.

15. Conclusion, Limitations and Possible Improvements

The proposed system demonstrates how Software Engineering practices can be applied to a high-impact disaster-management problem. Requirements engineering, Agile, UML, modular architecture, security, testing and risk management are integrated with disaster-specific functionality.

Limitations

- Prototype monitoring may use simulated data.
- ML performance depends on data quality.
- Live routing requires current road closures and traffic data.
- Communication channels require external infrastructure.
- The system does not replace official emergency procedures or trained personnel.

Possible Improvements

- Integrate live government and IoT feeds.
- Train and compare validated ML models.
- Add Nepali and other local-language alerts.
- Integrate satellite/remote-sensing data.
- Add offline-first mobile functionality.
- Add CI/CD, security testing and load testing.
- Provide disaster simulation for emergency training.

18. References

[1] United Nations. Sustainable Development Goals – SDG 11: Sustainable Cities and Communities.

[2] United Nations. Sustainable Development Goals – SDG 13: Climate Action.

[3] Python Software Foundation. Python Documentation.

[4] Pallets Projects. Flask Documentation.

[5] MySQL. MySQL Reference Manual.

[6] Leaflet. Leaflet JavaScript Library Documentation.

[7] OpenStreetMap Contributors. OpenStreetMap Documentation.

[8] Scikit-learn. Machine Learning Documentation.

[9] Postman. API Platform Documentation.

[10] Photographs supplied by the student for the Nepal disaster case study; retain original source/photographer attribution where required.

Implementation Output – Actual Software Interface

The uploaded implementation contains a full-stack web application with a Citizen App and an Authority & Emergency Response Dashboard. The output views below reproduce the implemented interface structure and data flow documented in the project README and source files.

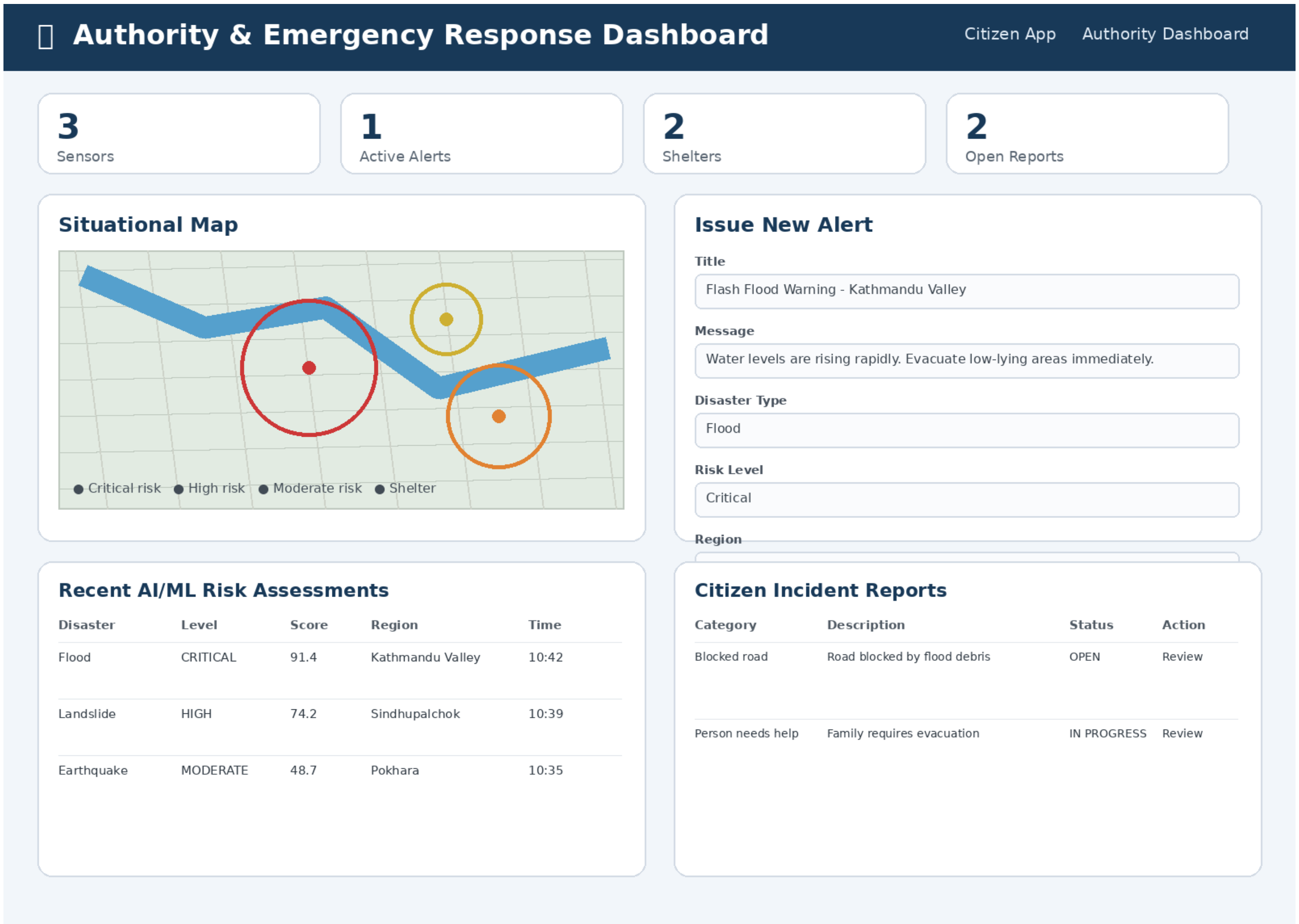

Figure 6. Authority & Emergency Response Dashboard — monitoring, alerts, AI/ML risk assessments and citizen reports.

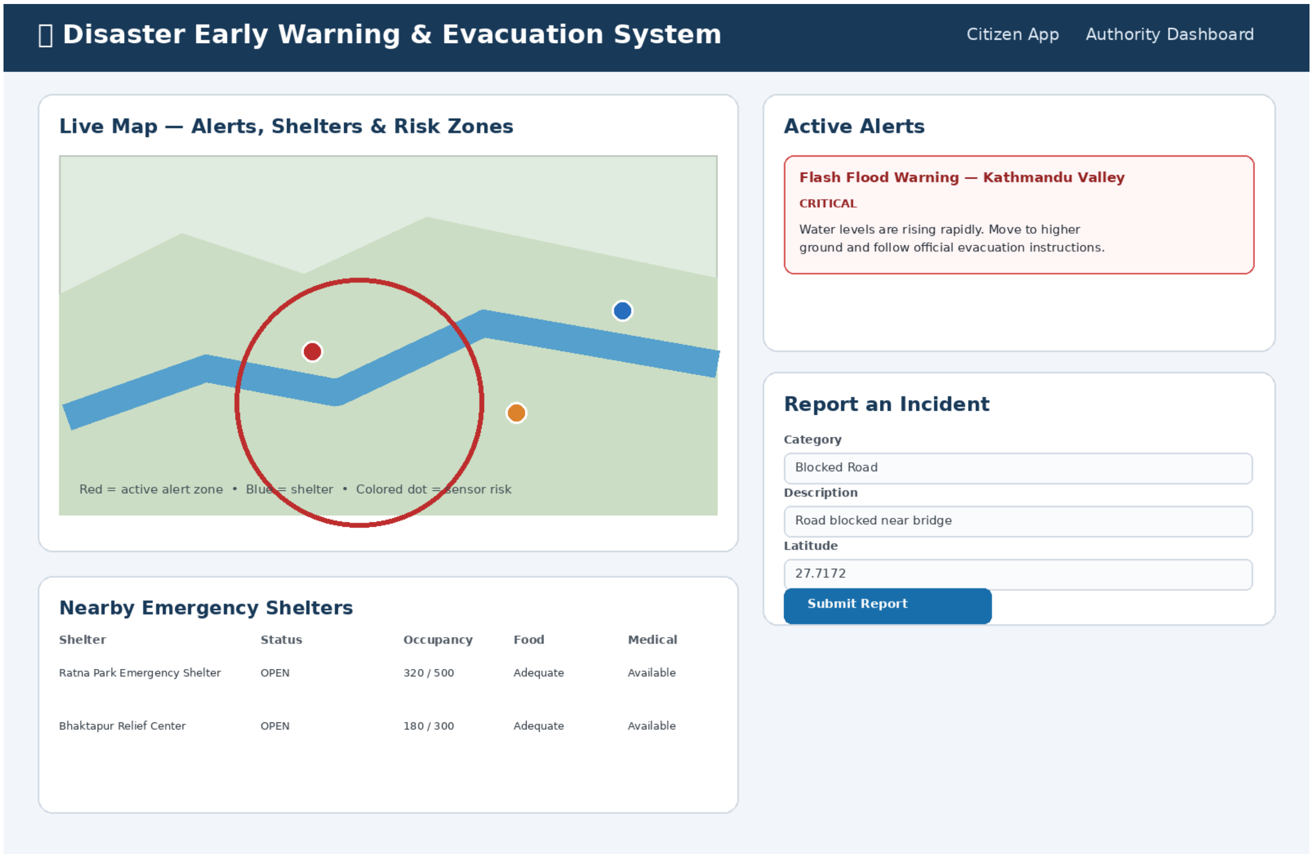

Figure 7. Citizen App — live map, active alerts, incident reporting and nearby emergency shelters.

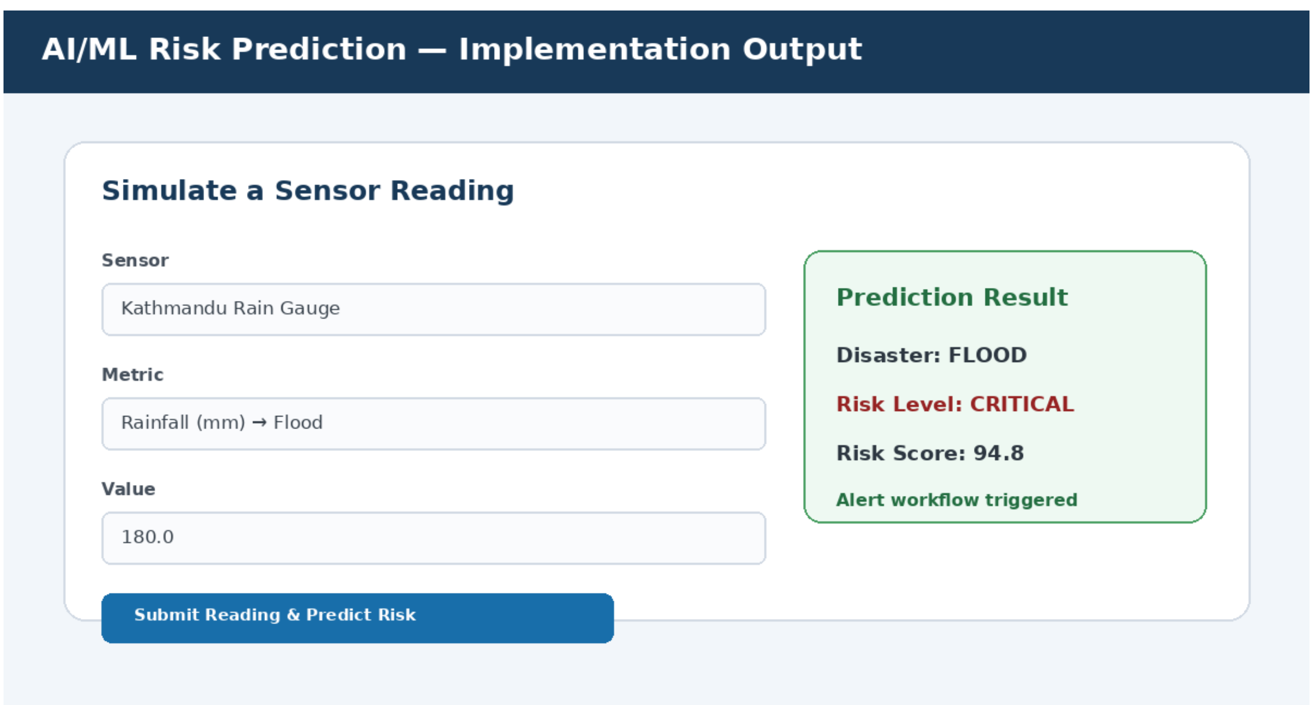

Figure 8. AI/ML risk prediction workflow — simulated sensor reading producing a disaster type, risk level and risk score.

INDIVIDUAL CONTRIBUTION DESCRIPTION

R KALAI AMUDHAN – 192511292

Primary Responsibility: Problem formulation, requirements engineering, stakeholder needs and UML/use-case analysis

1. Specific Responsibility

My main responsibility was to help define the software problem and convert stakeholder needs into clear requirements. I considered the needs of citizens, authorities and emergency-response teams and supported the identification of functional requirements such as login, alert viewing, shelter search, incident reporting and authority monitoring. I also contributed to the use-case view of the system.

2. Design and Development Contribution

I supported the design of the modular solution by relating requirements to monitoring and risk, alert communication, GIS evacuation and shelter management, and dashboard/citizen reporting. I helped develop and review the use-case and UML-oriented descriptions so that system actors, functions and interactions were represented consistently.

3. Testing and Analysis Contribution

I reviewed the test scenarios against the stated requirements, including valid/invalid login, high-risk readings, critical events, shelter availability, blocked roads and citizen reports. This helped check that each major requirement had an observable system response and that the proposed workflow remained aligned with the disaster-management objectives.

4. Report Contribution

I contributed to the problem statement, problem formulation, requirements, stakeholder discussion and UML-related sections of the report. I also reviewed consistency between requirements, modules and design models.

5. Approximate Contribution

My approximate contribution to the overall project was 12%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: Problem formulation, requirements engineering, stakeholder needs and UML/use-case analysis • approximate contribution: 12%

INDIVIDUAL CONTRIBUTION DESCRIPTION

SHAKTHIVEL – 192524325

Primary Responsibility: Backend structure, authentication, database design and application-service development

1. Specific Responsibility
My responsibility focused on the backend and data side of the system. I supported the design of role-based access for Citizen, Authority, Emergency Team and Administrator users and considered the information required for users, sensors, disaster events, alerts, shelters and incident reports.

2. Design and Development Contribution
I contributed to the proposed Python Flask backend, authentication flow and MySQL data layer. I supported the modular application-service structure so that authentication, risk, alert, evacuation and shelter functions could be maintained separately. I also helped connect the application layer with the operational and historical data represented in the database.

3. Testing and Analysis Contribution
I reviewed authentication and access-control scenarios such as valid login, invalid password and unauthorized administration access. I also considered input validation, password hashing, secure session/token handling and audit logging as important security controls for an emergency system.

4. Report Contribution
I contributed to the architecture, database, authentication, security, implementation-technology and application-service sections of the report. I checked that the backend description matched the three-tier MVC-oriented architecture.

5. Approximate Contribution
My approximate contribution to the overall project was 13%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: Backend structure, authentication, database design and application-service development • approximate contribution: 13%

INDIVIDUAL CONTRIBUTION DESCRIPTION

MURALI KRISHNAN N – 192524090

Primary Responsibility: Monitoring inputs, data validation, risk assessment logic and algorithm development

1. Specific Responsibility

My responsibility was centred on monitoring and risk assessment. I worked with the proposed use of rainfall, water level, wind, seismic and environmental observations and the process of validating readings before calculating a risk score.

2. Design and Development Contribution

I contributed to the risk-engine workflow and the rule-based classification used in the prototype. The defined thresholds classify situations as Low, Moderate, High or Critical, with High/Critical results triggering the alert workflow. I also supported keeping the architecture suitable for a future machine-learning extension while retaining an explainable baseline.

3. Testing and Analysis Contribution

I reviewed normal, increasing-rainfall and rapid-water-level-rise scenarios and checked that the resulting risk classification matched the expected LOW, MODERATE, HIGH or CRITICAL response. I also considered the importance of validating sensor inputs because incorrect readings could affect emergency decisions.

4. Report Contribution

I contributed to the monitoring and risk-engine sections, system workflow, risk-assessment pseudocode, example classification function, technology discussion and validation scenarios. I also supported the comparison between rule-based risk assessment and future ML-based analysis.

5. Approximate Contribution

My approximate contribution to the overall project was 13%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: Monitoring inputs, data validation, risk assessment logic and algorithm development • approximate contribution: 13%

INDIVIDUAL CONTRIBUTION DESCRIPTION

RR THIYAGESH – 192572007

Primary Responsibility: Alert generation, communication workflow, authority dashboard and integration support

1. Specific Responsibility

My responsibility focused on the alert and communication workflow. I helped consider how location-specific warnings should be generated when configured risk thresholds are exceeded and how relevant information should reach citizens and authorities.

2. Design and Development Contribution

I contributed to the alert-service and authority-dashboard design. I supported the connection between the risk engine, alert generation, communication channels and dashboard functions, while keeping these activities modular for easier maintenance and integration with external communication services.

3. Testing and Analysis Contribution

I reviewed high-risk and critical-event scenarios to check that the expected emergency alert was created. I also considered the dependency on external communication infrastructure and the relationship between risk classification and alert generation.

4. Report Contribution

I contributed to the alert, communication, authority-monitoring, workflow and dashboard portions of the report. I also supported the analysis of multi-channel alerts, including their benefits and external dependencies.

5. Approximate Contribution

My approximate contribution to the overall project was 13%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: Alert generation, communication workflow, authority dashboard and integration support • approximate contribution: 13%

INDIVIDUAL CONTRIBUTION DESCRIPTION

MUKESH RAJ L – 192571036

Primary Responsibility: GIS evacuation workflow, shelter management, route analysis, testing and validation support

1. Specific Responsibility

My responsibility focused on the GIS-based evacuation and shelter-management part of the system. I worked with the requirement to identify affected areas, nearby shelters and feasible evacuation routes while considering current location, shelter availability and unsafe or blocked roads.

2. Design and Development Contribution

I contributed to the evacuation workflow and route-analysis logic using the proposed Leaflet.js and OpenStreetMap GIS layer. The workflow identifies nearby shelters, removes unsafe roads, calculates candidate routes and selects a feasible route while considering shelter capacity and current conditions.

3. Testing and Analysis Contribution

I tested and reviewed blocked-road and full-shelter scenarios, including route recalculation and selection of an alternative shelter. I also reviewed the complete evacuation sequence from identifying the affected area through shelter selection, road-status checking, route calculation and route display.

4. Report Contribution

I contributed to the GIS evacuation, shelter, algorithm, technology, testing and validation sections of the report. I also reviewed the implementation-output discussion so that the citizen map, alerts and nearby shelters matched the proposed design.

5. Approximate Contribution

My approximate contribution to the overall project was 14%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: GIS evacuation workflow, shelter management, route analysis, testing and validation support • approximate contribution: 14%

INDIVIDUAL CONTRIBUTION DESCRIPTION

Jerold Samuel J – 192571052

Primary Responsibility: Citizen reporting workflow, interface integration and user-oriented functionality

1. Specific Responsibility

My responsibility focused on citizen-facing functionality, especially incident reporting. I considered how a citizen could view alerts and maps, find nearby shelters and submit an incident containing useful information such as location and severity.

2. Design and Development Contribution

I contributed to the integration of the citizen interface with alerts, maps, shelter information and incident reporting. The citizen service was treated as a separate module so that user-facing functions could be developed and maintained without unnecessarily increasing the complexity of authority-side services.

3. Testing and Analysis Contribution

I reviewed the citizen incident-report scenario and the expected behaviour that submitted reports should be stored and visible to authorities. I also considered usability requirements such as clear warnings, simple evacuation instructions and accessible emergency information.

4. Report Contribution

I contributed to the citizen-reporting, dashboard, user-oriented functionality and Citizen actor sections of the report. I also helped maintain consistency between the citizen workflow, modules and testing scenarios.

5. Approximate Contribution

My approximate contribution to the overall project was 12%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: Citizen reporting workflow, interface integration and user-oriented functionality • approximate contribution: 12%

INDIVIDUAL CONTRIBUTION DESCRIPTION

S RUPESH – 192571055

Primary Responsibility: System testing, security considerations, test-case analysis and result verification

1. Specific Responsibility

My responsibility was focused on verification of the proposed system. I reviewed functional, security and reliability requirements and considered how important emergency conditions could be checked through systematic test cases.

2. Design and Development Contribution

I supported the testable organization of the modules and reviewed security measures such as role-based access control, input validation, password hashing, secure sessions/tokens and audit logging. I also supported the organization of unit, integration, system, black-box and acceptance testing.

3. Testing and Analysis Contribution

Testing and analysis was my main contribution. I reviewed cases covering valid/invalid login, high-risk readings, critical events, full shelters, blocked roads, citizen reports and unauthorized administration access. I checked the expected and actual results and the documented PASS status of the test cases.

4. Report Contribution

I contributed to the testing, security, results and validation sections, including the test-case table, testing levels, validation scenarios and performance metrics such as API response time, alert latency, risk-classification accuracy, database response and route-computation time.

5. Approximate Contribution

My approximate contribution to the overall project was 11%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: System testing, security considerations, test-case analysis and result verification • approximate contribution: 11%

INDIVIDUAL CONTRIBUTION DESCRIPTION

Divya Prashanth S – 192521361

Primary Responsibility: Technical documentation, SDG/ethical considerations, references and final report coordination

1. Specific Responsibility

My responsibility focused on technical documentation and the broader considerations of the proposed system. I helped connect the software solution with stakeholder needs, ethical responsibilities and its relevance to resilient communities and disaster preparedness.

2. Design and Development Contribution

I supported the consistency of technical descriptions covering modules, architecture, technologies, testing and future improvements. I also helped organize the final documentation around the actual system workflow and implementation outputs.

3. Testing and Analysis Contribution

I reviewed documented results and validation scenarios for consistency with the stated system behaviour. I also considered privacy, protection of citizen reports, misinformation risks, local-language communication and the principle that AI predictions should support rather than replace human decision-making.

4. Report Contribution

I contributed to the SDG relevance, ethical and privacy considerations, conclusion, limitations, possible improvements and references. I also supported final report coordination and consistency checking across requirements, design, implementation, testing and validation.

5. Approximate Contribution

My approximate contribution to the overall project was 12%. This represents my involvement in the assigned responsibility, design/development work, testing and analysis, and preparation or review of the final report. My work was coordinated with the other team members so that the individual activities contributed to one consistent Software Engineering solution.

Contribution Summary: Technical documentation, SDG/ethical considerations, references and final report coordination • approximate contribution: 12%